Assignment 2

Electricity and Electronics EBN 111

11 March 2024

Test Information			
Maximum marks:	20	Full marks:	20
Due Date:	4 March 2024 (23:59)	Open/Closed book:	Open
Total number of pages (this page included)		12	

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Important

1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers, including complex numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your assessment, you declare the following:**

   The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

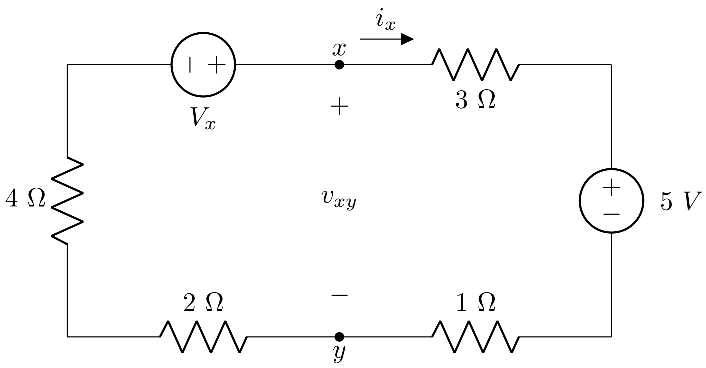
   I hereby declare that the work herewith is completely my own and that no parts have been copied in any way from current or previous students or other sources.
6. Contact ebn111.2024@gmail.com **before** the deadline for assistance with technical difficulties.

Additional guide for Complex numbers

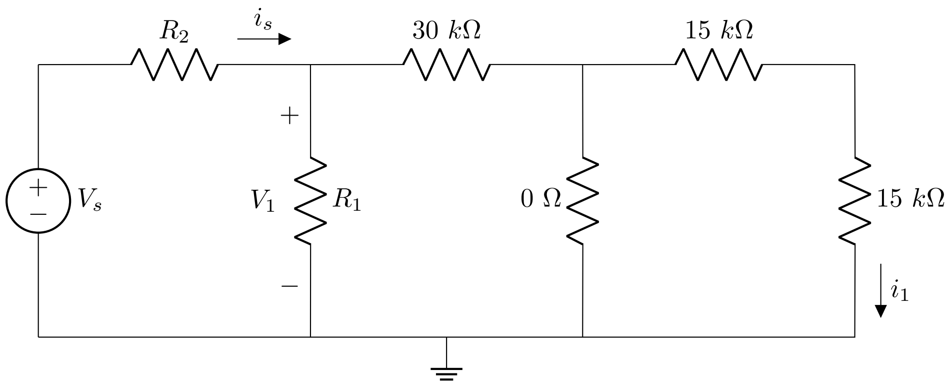
The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

- **Correct**: *1-j2* or *-3+j4*
  - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
  - Capital J is however incorrect, and *1-2j* is also not acceptable.
  - If a complex number has no real part, for example *0+j2*, write *0+j2* as the final answer.
- **Correct**:
  - With angle in degrees: $12\∠34\°$
  - With angle in radians: $12\∠0.5934$
  - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect**:
  - *1-i2* (Don't use i, use j)
  - *1 - j2* (Don't use spaces between characters)
  - $12\ \∠\ 34\ \°$ (Don't use spaces between characters)
  - *j10+1* (Don't write the imaginary part before the real part for rectangular form.)
  - *-3+4j* (Don't write the j after the number for imaginary numbers, write $j4$ instead of $4j$)

Section 1

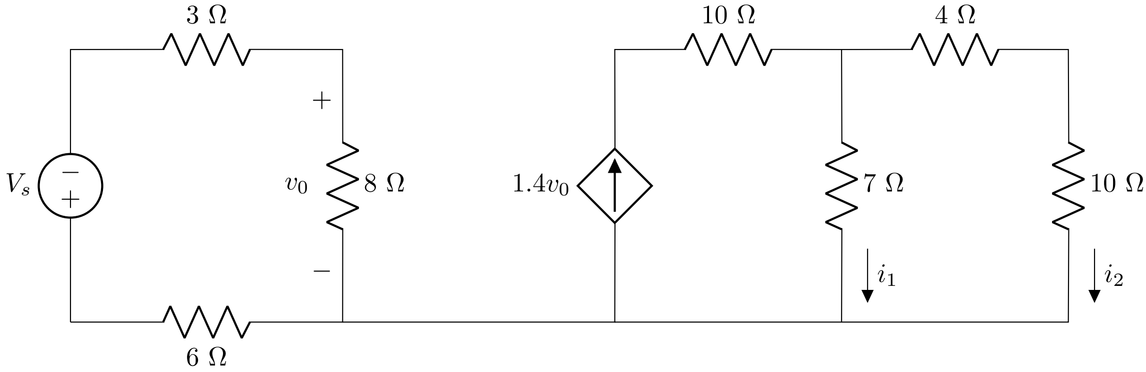
Question 1 [2]	
Given: V_x	
	
01	Determine v_{xy} [2]

Question 2 [2]
Given: $V_s,\ R_1,\ R_2$



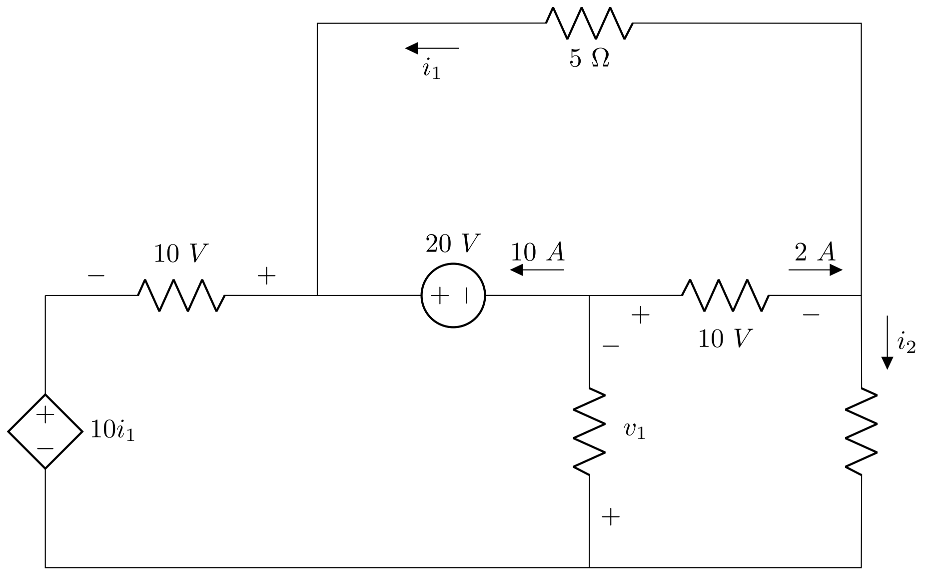
02	Find V_1 [2]

Question 3 [1]	
Given: V_s	



03	Determine v_o. [1]

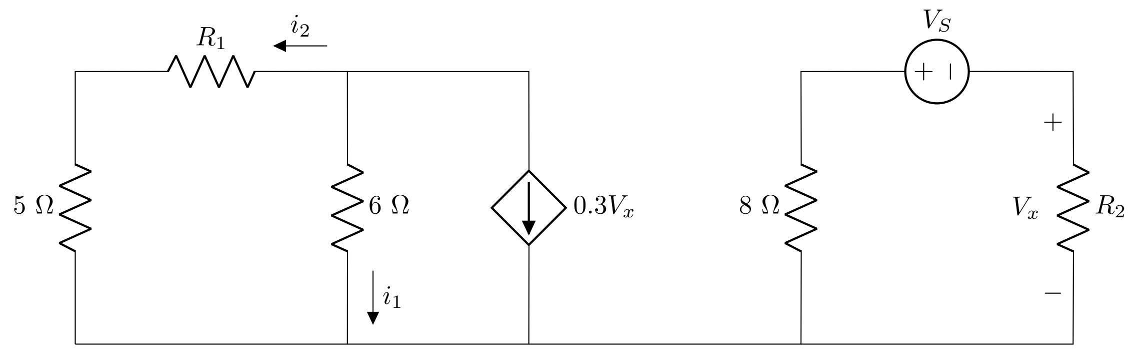
Question 4 [1]	
Given: i_1.	



04	Determine i_2. [1]

Questions 5 to 6 [3]

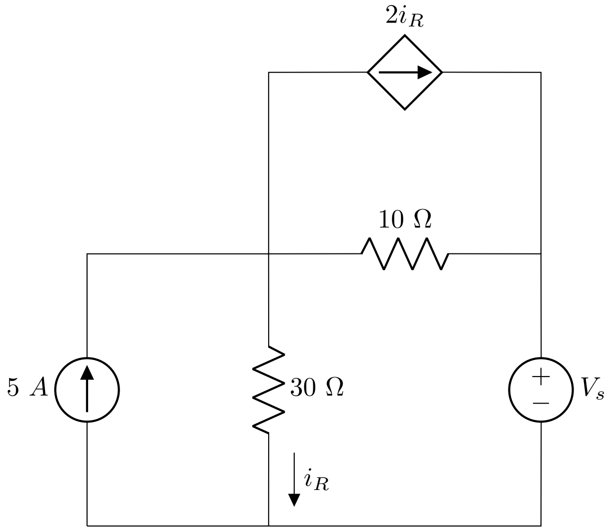
Given: $V_S,\ R_1,\ R_2$.



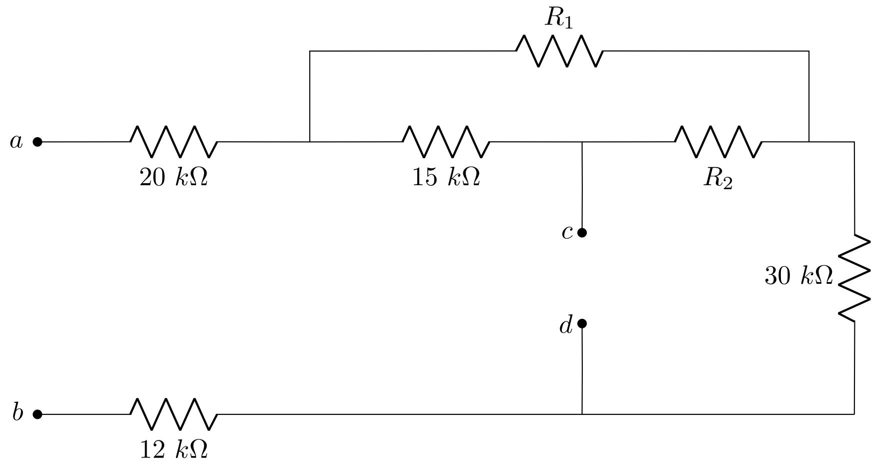
05	Determine the power P_1 dissipated in R_2. [2]
06	Determine i_1. [1]

Question 7 [2]

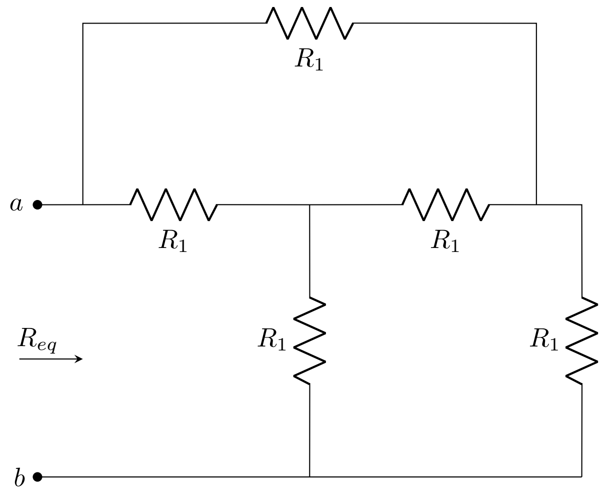
Given: i_R



07 Calculate V_s [2]

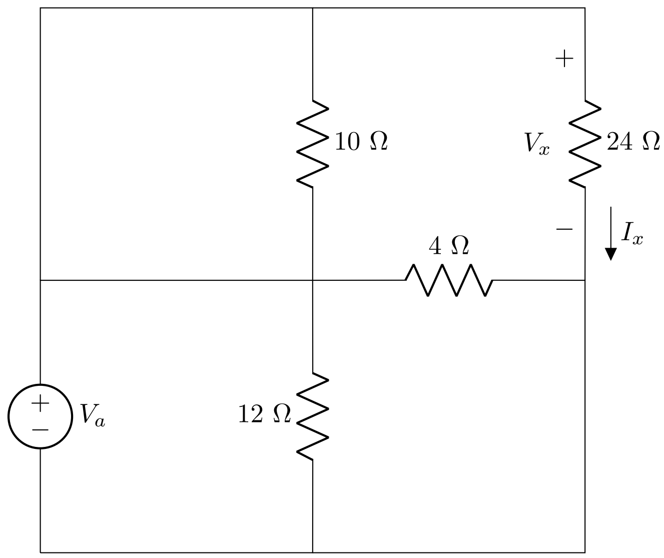
<table>
<tr><td colspan="2">Question 8 [2]</td></tr>
<tr><td colspan="2">Given: R_1, R_2</td></tr>
<tr><td colspan="2">
</td></tr>
<tr><td>08</td><td>Calculate the equivalent resistance R_{cd} with respect to terminals c-d. [2]</td></tr>
</table>

Question 9 [2]	
Given: R_1	

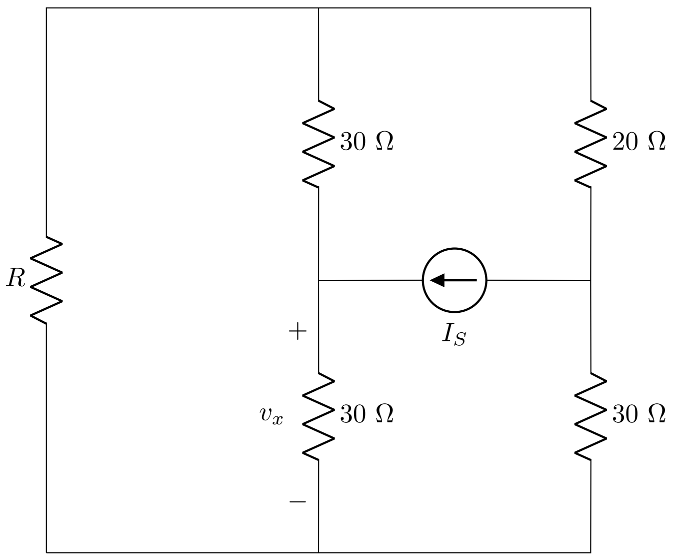


09	Calculate R_{eq} [2]

Questions 10 to 12 [3]	
How many nodes, independent loops, and branches are there in the following circuit?	



10	Nodes [1]
11	Loops [1]
12	Branches [1]

Question 13 [2]	
Given: $R = \infty$ and $I_S = [I_S]A$	
	
13	Calculate v_x. [2]

Total Section 1: [20]

Grand Total : [20]